Week 10 - Graded Assignment 10

The due date for submitting this assignment has passed.
Due on 2026-04-22, 23:59 IST.
You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-04-22, 07:58 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

1 point
Which of the following is (are) the critical points of the scalar valued function $f(x,y) = 3x^2y + y^3 - 3x^2 - 3y^2 + 2$f(x, y) = 3x2y + y3 − 3x2 − 3y2 + 2?

- [x] $(0,0)$(0, 0)
- [x] $(0,2)$(0, 2)
- [x] $(1,1)$(1, 1)
- [ ] $(1,2)$(1, 2)
- [ ] $(0,1)$(0, 1)

Yes, the answer is correct.
Score: 1
Accepted Answers:
$(0,0)$(0, 0)
$(0,2)$(0, 2)
$(1,1)$(1, 1)

1 point
Consider a function $f : R^2 \to R$f : R2 → R defined as:

$$f(x,y) = \begin{cases} \frac{xy^2}{x^2+y^4}, & \text{if } (x,y) \neq (0,0) \\ 0, & \text{otherwise} \end{cases}$$
f(x, y) = {x2+y4xy2,0,if (x, y) = (0, 0)otherwise

Which of the following is (are) true about $f(x,y)$f(x, y)?

- [ ] The directional derivative at $(0,0)$(0, 0) in the direction of a unit vector $u = (u_1, u_2)$u = (u1, u2) is 1.
- [ ] The directional derivative at $(0,0)$(0, 0) in the direction of a unit vector $u = (u_1, u_2)$u = (u1, u2) is $\frac{u_2^2}{u_1}$u1 u22, where u_1 u1 is non-zero.
- [ ] Amongst all directional derivatives at $(0,0)$(0, 0), the maximum occurs in the direction of the vector $(5,5)$(5, 5).
- [x] There is no plane which contains all the tangent lines at $(0,0)$(0, 0) and hence the tangent plane)(0, 0) does not exist.

Partially Correct.
Score: 0.5
Accepted Answers:

The directional derivative at $(0,0)$ in the direction of a unit vector $u = (u_1, u_2)$ is $\frac{u_2^2}{u_1}$, where u_1 is non-zero.
There is no plane which contains all the tangent lines at $(0,0)$ and hence the tangent plane at $(0,0)$ does not exist.

1 point

Let $f : R^3 \to R$ be the function defined by $f(x, y, z) = 2x^3 + y^2 - z^3$, for all $(x, y, z) \in R^3$. Choose all the directions along which there is no change in the function f at the point $(1, 1, 1)$.

- [x] $(\frac{1}{\sqrt{5}}, 0, \frac{2}{\sqrt{5}})$
- [ ] $(\frac{6}{7}, \frac{2}{7}, -\frac{3}{7})$
- [ ] $(\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}})$
- [x] $(0, \frac{3}{\sqrt{13}}, \frac{2}{\sqrt{13}})$
- [x] $(\frac{2}{\sqrt{17}}, -\frac{3}{\sqrt{17}}, \frac{2}{\sqrt{17}})$

Yes, the answer is correct.
Score: 1
Accepted Answers:

$(\frac{1}{\sqrt{5}}, 0, \frac{2}{\sqrt{5}})$

√
2)
$(0, \frac{3}{\sqrt{13}}, \frac{2}{\sqrt{13}})$(0, 13
√
3, 13
√
2)
$(\frac{2}{\sqrt{17}}, -\frac{3}{\sqrt{17}}, \frac{2}{\sqrt{17}})$(17
√
2, − 17
√
3, 17
√
2)

Let $L_f(x, y) = Ax + By + C$Lf(x, y) = Ax + By + C be the linear approximation to the function $f(x, y) = ye^x - \frac{1}{4}(x^2 + y^2)$f(x, y) = yex − 41(x2 + y2) at (0, 1)(0, 1). Then find the value of $A + 2B + 4C$A + 2B + 4C.
3
Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 3
1 point
Consider a function $f(x, y) = 2\sqrt{x^2 + 4y}$f(x, y) = 2 x2 + 4y
√
. Let SS denote the set of unit vectors uu for which the directional derivative of ff at $(-2, 3)$(−2, 3) in the direction of uu is 00. Find the cardinality of the set SS.
2
Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 2
1 point
Suppose $f(x, y) = xye^x$ f (x, y) = xyex be a scalar valued multivariable function. Then using the linear approximation $L_f(x, y)$Lf(x, y) of the function ff at (1, 1)(1, 1), the estimate value of $f(1.2, 0.9)$f(1.2, 0.9) is found to be βeβe, where ββ is a real number. The value of ββ is
1.3
Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Range) 1.25,1.35
1 point

The temperature TT (in degree centigrade, ^0C0C) in a solid metal sphere is given by the function $e^{-(x^2 + y^2 + z^2)}$ e−(x2+y2+z2). Answer Questions 6,7 and 8 from the given information.
1 point
Choose the set of correct options.
☐

The rate of change of temperature in the direction of X-axis is continuous at every point.

[ ] The rate of change of temperature in the direction of Z-axis is not continuous at the origin.

[x] The rate of change of temperature at the origin from any direction is constant and that is 0.

[ ] The rate of change of temperature at the origin from any direction is constant and that is e.

[ ] The rate of change of temperature at the origin from any direction is not constant.

Partially Correct.

Score: 0.5

Accepted Answers:

The rate of change of temperature in the direction of X-axis is continuous at every point.

The rate of change of temperature at the origin from any direction is constant and that is 0.

The rate of change of the temperature at point $(1, 0, 0)$ in the direction toward point $(8, 6, 0)$ is A. Find the value of $10Ae$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -16

1 point

1 point

Which of the following statements are true?

[x] At a point (a, b, c) on the sphere the maximum rate of change in temperature is given by $2e^{-(a^2+b^2+c^2)}\sqrt{a^2+b^2+c^2}$.

[ ] At a point (a, b, c) on the sphere the maximum rate of change in temperature is given by $-2e^{-(a^2+b^2+c^2)}\sqrt{a^2+b^2+c^2}$.

[x] At a point (a, b, c) on the sphere the maximum rate of change in temperature is in the direction of the unit vector $\left(-\frac{a}{\sqrt{a^2+b^2+c^2}}, -\frac{b}{\sqrt{a^2+b^2+c^2}}, -\frac{c}{\sqrt{a^2+b^2+c^2}}\right)$.

[ ] At a point (a, b, c) on the sphere the maximum rate of change in temperature is in the direction of the unit vector $\left(-\frac{2a}{e^{a^2+b^2+c^2}}, -\frac{2b}{e^{a^2+b^2+c^2}}, -\frac{2c}{e^{a^2+b^2+c^2}}\right)$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

At a point (a, b, c) on the sphere the maximum rate of change in temperature is given by

$2e^{-(a^2+b^2+c^2)}$ 2e−(a2+b2+c2) $\sqrt{a^2+b^2+c^2}$ a2 + b2 + c2

√

.

At a point (a,b,c)(a, b, c) on the sphere the maximum rate of change in temperature is in the direction of the unit vector $(-\frac{a}{\sqrt{a^2+b^2+c^2}}, -\frac{b}{\sqrt{a^2+b^2+c^2}}, -\frac{c}{\sqrt{a^2+b^2+c^2}})$(− a2+b2+c2

√

a, − a2+b2+c2

√

b, − a2+b2+c2

√

c).

Find out the maximum directional derivative at (0,0) of the function $f_1(x,y) = y^4 e^{6x}$ f1 (x, y) = y4e6x.

0

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0

1 point

Find out the maximum directional derivative at (0,0) of the function $f_2(x,y) = 5 - 6x^2 + 5x - 3y^2$ f2 (x, y) = 5 − 6x2 + 5x − 3y2.

5

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 5

1 point

Find out the maximum directional derivative at (0,0) of the function $f_3(x,y) = 6x\sin(6x) + 6y\cos(6y)$f3(x, y) = 6x sin(6x) + 6y cos(6y).

6

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 6

1 point

The equation of the tangent plane to the surface $z = 3 - x^5 - y^2$ z = 3 − x5 − y2 at the point (1, 1, 1) (1, 1, 1) is $z = Ax + By + C$z = Ax + By + C. Find the value of $C - A - B$C − A − B.

15

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 15

1 point

Let $f : R^2 \to R$f : R2 → R be a function such that the partial derivatives of ff exist and are continuous at every point. Suppose $f(1,1) = 3$f(1, 1) = 3, and (0, 1, 5)(0, 1, 5) is a point on the tangent line to the graph of f at $(1, 1)$ in the direction $(1, 0)$. Find $f_x(1, 1)$, the partial derivative with respect to x at $(1, 1)$.

-2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) -2

1 point